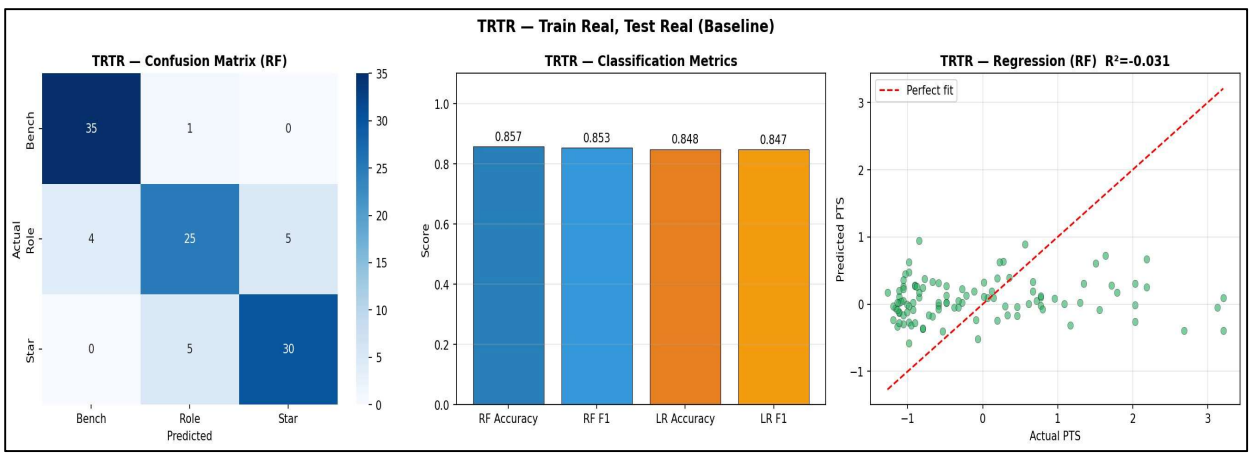
CSE 641 CV Project Report Week-7
Project Name: Athletic Readiness: Data Generation
Group Name: Deep Vision

Objective

This week our aim was to validate the quality of the synthetic data generated by the VAE and GAN using four testing strategies: TRTR, TSTR, TRTS, and TSTS. A classification model (predicting player tier) and a regression model (predicting PTS) were trained under each strategy using Random Forest and Logistic Regression / Ridge models. The results were compared against the TRTR baseline to determine how well each generative model can substitute real data.

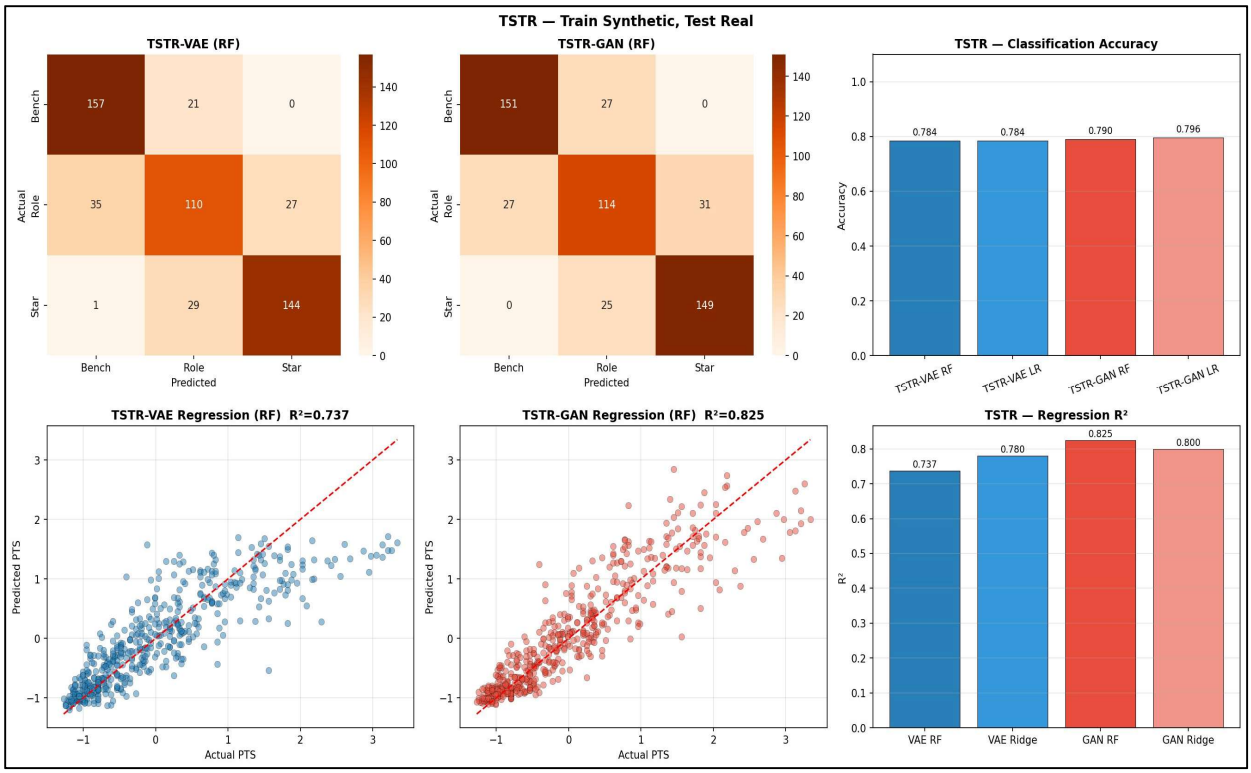
Strategy 1: TRTR — Train Real, Test Real (Baseline)

This defines the absolute limit of what is learnable from your specific dataset. If the real world has inherent noise and an algorithm can only learn it to 85.7% accuracy as the random forest classifier achieved an accuracy of 0.857 and F1 of 0.853 and confirms a strong and learnable signal in the real dataset. So, we cannot expect synthetic data to magically produce a 95% accuracy model. This establishes a target that the generative models are striving to match.



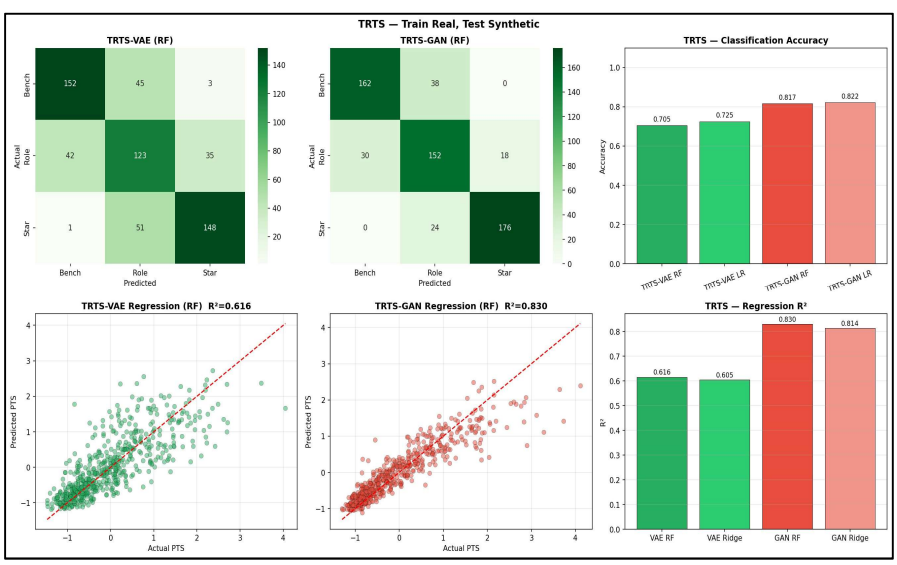
Strategy 2: TSTR — Train Synthetic, Test Real

This is the most crucial test as here athlete is training on a virtual reality simulation, and then sending him to play in a real physical game. Because algorithms trained on the synthetic data performed well in reality. So, it proves that simulations successfully captured the core, actionable mechanics of the real world. Models were trained both the VAE and GAN synthetic data and then tested on the full real dataset. Both the models performed close to the TRTR baseline on classification — VAE RF achieved 0.784 accuracy and GAN RF achieved 0.790, a gap of only ~0.07 from the baseline. For regression, the GAN outperformed the VAE with an R^2 of 0.825 vs 0.737. GAN generated data better captured the regression structure of the real data.



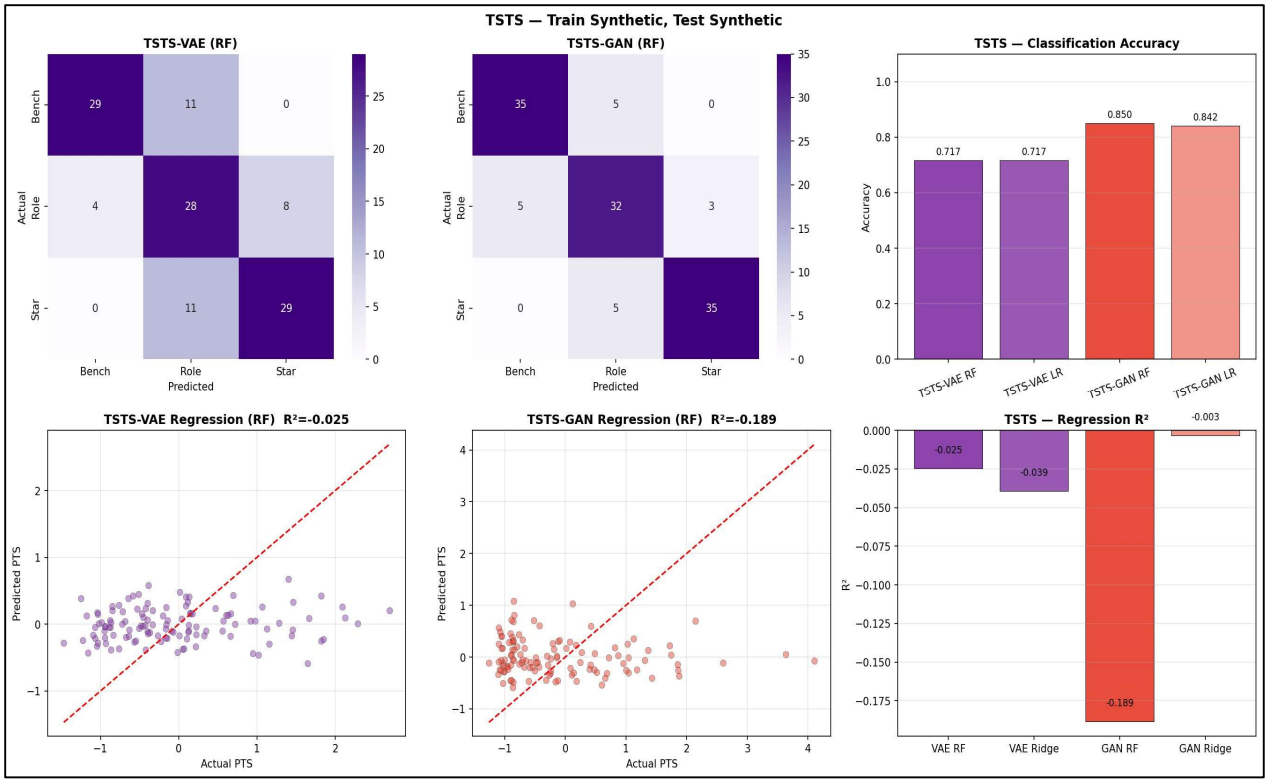
Strategy 3: TRTS — Train Real, Test Synthetic

We take an algorithm trained entirely on real-world physics and then test it in the simulation. If the model scores high, then it means that the simulated data "looks and feels" exactly like the real world to an algorithm that knows the reality. The GAN is clearly a much better statistical "mimic" of the real-world data distribution. Models trained on real data were tested on synthetic data. The synthetic data generate by the GAN scored significantly higher than the VAE — GAN RF accuracy of 0.817 vs VAE RF accuracy of 0.705, and GAN R^2 of 0.830 vs VAE R^2 of 0.616. This indicates that the GAN generated data more faithfully mirrors the statistical patterns of the real dataset, and a model trained on real data generalises better to GAN synthetic samples.



Strategy 4: TSTS — Train Synthetic, Test Synthetic

This test will show that if the synthetic universe has its own coherent rules or not. The results revealed a specific limitation: while the broad categories (player tiers) are structured and internally consistent, the nuanced continuous relationships (predicting exact points) dissolved into noise during the generative process. The synthetic space works well for big-picture classification, but loses structural fidelity on micro-level continuous targets. Models were trained and tested on the same synthetic dataset with the 80-20 split. The GAN achieved a higher internal classification accuracy of 0.850 compared to the VAE 0.717, this indicated that the GAN data is more internally structured and consistent. However, both models showed near-zero or negative R^2 scores for regression, which is expected since the PTS feature has very low predictive signal within the standardised synthetic-only space when other features are used.



Summary: All Strategies — Classification Accuracy (RF)

Strategy	Model	RF Accuracy	RF F1	RF R^2
TRTR (Baseline)	**Real → Real**	**0.857**	**0.853**	**-0.031**
TSTR	VAE → Real	0.784	0.781	0.737
TSTR	GAN → Real	0.790	0.788	0.825
TRTS	Real → VAE	0.705	0.708	0.616
TRTS	Real → GAN	0.817	0.818	0.830
TSTS	VAE → VAE	0.717	0.723	-0.025
TSTS	GAN → GAN	0.850	0.851	-0.189

Results & Observations

- Both the VAE and GAN synthetic data produced TSTR classification accuracy within ~7% of the TRTR baseline (0.857), which confirms that the synthetic data is a viable substitute for the real data.
- The GAN outperformed the VAE in TSTR regression (R^2: 0.825 vs 0.737). This indicates that the GAN data preserving the predictive structure of real data better.
- In TRTS, the GAN scored significantly higher than the VAE (Acc: 0.817 vs 0.705; R^2: 0.830 vs 0.616), this confirms that the GAN synthetic samples are closely resembling the real data distribution.
- In TSTS, the negative R^2 values for regression in both models indicated that the PTS cannot be reliably predicted from other features within the synthetic space only. It is likely due to the noise in the generative process for the continuous targets.
- Overall, the GANs outperformed the VAEs across most evaluation strategies, particularly in TRTS and regression tasks.

Conclusion

The four testing strategies confirmed that both the VAE and GAN successfully generate synthetic data that is useful for downstream machine learning tasks. The TSTR classification accuracy of ~0.79 for both models against the 0.857 baseline demonstrates strong synthetic data quality. The GAN consistently outperformed the VAE in cross-domain testing (TSTR, TRTS), making it the stronger generative model for this dataset. This completes the implementation and evaluation phase of the Athletic Readiness synthetic data generation project.